

SUMMARY

Computer Science senior building distributed and real-time systems. Creator of *wikiracing.org* (4,200+ monthly active users; 80,000+ game starts) and *CodeCafé* (500+ GitHub stars); Apache Cassandra contributor.

EDUCATION

- Gettysburg College

Bachelor of Science in Computer Science; Minor in Economics; Major GPA: 3.63/4.0

Honors: Dean’s List (2023, 2024); Presidential Scholar

Coursework: Computer Networks, Artificial Intelligence, OOP, Data Structures & Algorithms, Databases, Comp Org

Gettysburg, PA

Expected Dec 2026

PROJECTS

- wikiracing.org — Real-Time Multiplayer Wikipedia Racing Platform (4,200+ MAU)

Live · Gameplay

Dec 2025 – Present

Creator & Sole Maintainer

Concurrent WebSocket Backend: Built the Go WebSocket backend behind 80,000+ game starts with per-client read/write goroutines, bounded send queues, heartbeats, and token-bucket rate limits; mutex-guarded rooms coordinate matchmaking, parties, live positions, and reconnects.

Durable, Verified Results: Built idempotent result ingestion with per-account and guest outboxes, JWT and opaque guest authentication, and a background Go worker that validates each claimed hop against Wikipedia before crediting PostgreSQL player statistics.

Ghost Replay System: Records cursor and navigation events as timestamped streams; a Go service validates and promotes candidates to Cloudflare R2, then serves them as synchronized ghost opponents.
- CodeCafé — Collaborative Cloud IDE (500+ ★, Hacker News front page)

GitHub · Live · Walkthrough

Jan 2025 – Present

Creator & Sole Maintainer

Custom OT Engine (Core): Wrote a conflict-free Operational Transformation (OT) engine from scratch (like Google Docs); it reconciles simultaneous edits across users in real time using Java Spring Boot, WebSockets, and Redis Lua scripts.

Cloud IDE Architecture: Built a browser-based workspace using React, TypeScript, and Monaco Editor that gives users an instant live preview of their HTML/CSS/JS code.

Infrastructure & CI/CD: Dockerized the app and wired GitHub Actions to run Jest/JUnit tests and deploy the backend to AWS EC2 and the front end to Vercel; Nginx terminates TLS and proxies WebSockets, with Cloudflare for DNS/CDN.
- Gedis — Redis Server Built from Scratch in Go

GitHub

July 2025 – Present

Creator & Lead Developer

Storage Engine & Replication: Built a concurrent 30-command RESP2 server in Go with binary-safe pipelining, typed keys, TTLs, and skip-list-indexed sorted sets; the race-tested implementation adds crash-recoverable AOF persistence, atomic rewrites, and full/partial primary-replica resynchronization via replication IDs, offsets, and a backlog.

OPEN SOURCE

- Apache Cassandra

View Merged Contribution

Aug 2025 – Present

Open Source Contributor

Shipped in Cassandra 5.0.6 & 6.0: Made complex settings machine-readable as JSON in the settings virtual table so external tools can parse them without custom logic; preserved existing 5.0 behavior behind a system property. Reviewed by two Apache committers.

EXPERIENCE

- Software Engineer

Remote

Oct 2025 – Mar 2026

Poozle — AI Job-Matching App

Vector Recommendation Engine: Built a Rocchio-based filter on Pinecone that learns from rejections, aggregating embeddings of jobs a user dismissed and suppressing semantically similar listings from later results.

Streaks Progress Tracking: Shipped a streak system across React Native and Node.js that derives progress lazily from event timestamps on read, rather than updating counters in a nightly batch job.
- Software Engineering Intern

Gettysburg, PA

May 2025 – Aug 2025

Gettysburg College

Custom MCP Server & AI Tutor: Built a Cursor-style coding tutor into DrJava, the editor used in our intro CS courses. Wrote its MCP server from scratch on Java’s built-in HTTP server (no frameworks); file-read and directory tools ground the model in the student’s real code, streamed over SSE into a Swing panel.

Academic Integrity Platform: Created a React app for proctored coding exams that tracks keystrokes, monitors window focus, and builds a timeline of student activity.

PROGRAMMING SKILLS

Languages	Java · Go · JavaScript · TypeScript · Python · Lua · SQL
Frameworks & Tools	React · Node.js · Spring Boot · FastAPI · React Native · JUnit · Docker · GitHub Actions
Databases & Storage	PostgreSQL · MySQL · Cassandra · Redis · Amazon S3 · ElastiCache
Concepts	Distributed Systems · Event-Driven Architectures · WebSockets · REST · gRPC